

INTERPRETING NEGATION UNDER *Yiqian* ‘BEFORE’ IN MANDARIN CHINESE*

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Qualifying Paper 1

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Abstract

This paper investigates the puzzle of ‘expletive negation’ in Mandarin Chinese temporal clauses introduced by *yiqian* ‘before’, where the presence of negation often fails to alter the sentence’s truth conditions. Standard semantic analyses of ‘before’, which typically rely on selecting the earliest time point of the embedded proposition (e.g., Beaver & Condoravdi 2003 and Rett 2020), fail to account for the Mandarin data. To solve the puzzle, I build on Lin 2016 and propose a novel formal analysis where *yiqian* selects for a *transition* operator, which targets the unique non-infinite boundary of a temporal set. Further, I integrate this operator into a two-level aspectual coercion system, arguing that the interaction between aspect and negation is governed by the ontological distinction between states and events, following Kusumoto 2011. The proposed formal system correctly derives the apparent semantic vacuity of negation in embedded eventive predicates under *yiqian*, while predicting that negation remains interpreted in embedded stative predicates. Finally, I discuss how *yihou* ‘after’ might fit into the proposed formal mechanisms. The paper concludes with a methodological discussion on expletive negation from a cross-linguistic perspective and avenues for future research.

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1 Introduction

Cross-linguistically, there exists a phenomenon where the negation embedded under certain operators does not alter the truth conditions of the sentence, as if the negation is semantically vacuous. This phenomenon is commonly dubbed ‘expletive negation’ in the literature and has been discussed extensively within Romance languages.¹ The phenomenon is also observed in Mandarin Chinese and named by scholars ‘redundant negation’ (e.g., Lü 1979, Lien 1979 and Xiao & McEnery 2008), but few formal analyses for Mandarin expletive negation have been offered in the literature.²

The main concern of the current study is expletive negation embedded under *yiqian* ‘before’ as in a construction with the form ‘*p yiqian, q*’, meaning ‘*q* before *p*’:

- (1) a. yisheng lai yiqian, bingren jiu yijing si le.
 doctor come before patient then already die PFV.
 ‘Before the doctor arrived, the patient had already died.’
 b. yisheng **mei** lai yiqian, bingren jiu yijing si le.
 doctor NEG come before patient then already die PFV.
 ‘Before the doctor arrived, the patient had already died.’ (Adapted from Lin 2016)

The negation embedded under the *yiqian*-clause makes no difference to the truth conditions of the sentence, and therefore (1-a) and (1-b) have the exact same truth conditions. Lin 2016 proposes an explanation for this observation by arguing that ‘before’ across languages are sensitive to different operators that select a certain time point in the interval denoted by the embedded proposition.

It is worth noting that ‘before’ is a common trigger of expletive negation cross-linguistically, manifested in genetically unrelated languages alongside Mandarin (Jin & Koenig 2020), as exemplified below by the *abans*-clause in Catalan, the *prima*-clause in Italian, and the *bevor*-clause in German:

- (2) Haurem d’intervenir **abans** que (**no**) arribi el nou gerent.
 have.FUT to.take.part **before** that (NEG) arrive the new manager
 We’ll take part in the discussion before the new manager arrives. (Catalan, Espinal 1992)
 (3) Lo fermerai **prima** che (**non**) faccia qualche sciocchezza.
 him stop.2.FUT **before** that (NEG) do.3.SUBJ some folly
 You will stop him before he does anything silly. (Italian, Del Prete 2008)
 (4) Peter wollte Potsdam nicht verlassen **bevor** (**nicht**) das Projekt in
 Peter want.3SG.PST Potsdam NEG leave.INF before NEG DET project in
 ruhigem Fahrwasser war.
 calm.DAT.NEUT fairways be.3SG.PST
 ‘Peter did not want to leave Potsdam **before** the project was running smoothly.’
 (German, Krifka 2012; glosses added by current author)

However, the licensing conditions of expletive negation under ‘before’ are different across languages. In Catalan, expletive negation under *abans* is restricted to the non-factual (in some litera-

¹See Delfitto 2020 for an overview of the phenomenon and Jin & Koenig 2020 for a cross-linguistic survey of expletive negation triggers.

²For consistency in terminology, the current paper will use ‘expletive negation’ to refer to semantically vacuous negation from a descriptive perspective.

ture non-veridical) reading of the *abans*-clause, as is discussed in Espinal 2000 and Krifka 2012; in Italian, expletive negation under *prima* is restricted to anti-veridical (in some literature ‘counter-factual’) *prima*-clauses (Del Prete 2008); in German, expletive negation under *bevor* only makes sense if there is a negation in the matrix clause (Krifka 2012).

In contrast, expletive negation under *yiqian* in Mandarin occurs freely and appears unrestricted. As exemplified below, negation under *yiqian* is semantically vacuous in factual, non-factual and anti-veridical readings of *yiqian*-clauses:³

- (5) wo (mei) chusheng yiqian, wo jia jiu you fangzi le.
 I NEG born before I family then have house PFV
 My family had owned a house before I was born.
- (6) Zhangsan (mei) dao chaoshi qian, yudao le ta de jiaoshou.
 Zhangsan NEG arrive supermarket before meet PFV he POSS professor
 Zhangsan met his professor before he arrived at the supermarket.
- (7) zhadan (mei) baozha yiqian, jiu bei chaichu le.
 bomb NEG explode before then PASS remove PFV
 Bombs were defused before they exploded. (Adapted from Lin 2016)

(5) is considered to be only compatible with the factual reading because ‘I was born’ must be true in order for (5) to be uttered. (6) has a non-factual (non-committal) reading as after Zhangsan met his professor on his way to the supermarket, he could either continue his trip and finally arrive at the supermarket or abandon his plan to go to the supermarket and go somewhere else. The *yiqian*-clause in (7) only has an anti-veridical reading because it must be false that the bombs exploded. The negation under *yiqian* is semantically vacuous in all these three readings of *yiqian*-clauses, unlike Catalan and Italian where the licensing of expletive negation under ‘before’ is restricted.

As also shown by the above examples, unlike German, Mandarin expletive negation under *yiqian* does not require a co-occurring matrix negation. When there is a negation in the matrix clause, the matrix negation is interpreted and the embedded negation under ‘before’ remains semantically vacuous, i.e., the polarity of the matrix clause does not have any effect on the interpretation of the *yiqian*-clause, as exemplified by the following example:

- (8) wo (mei) chusheng yiqian, wo jia mei you fangzi.
 I NEG born before I family NEG have house
 My family did not own a house before I was born.

The cross-linguistic variation in the distribution of expletive negation under ‘before’ suggests that it is possible that the *prima facie* same phenomenon actually has different underlying formal mechanisms across languages and calls for different analyses.

The current paper focuses on expletive negation under *yiqian* in Mandarin but also includes discussions on various expletive negation triggers from a cross-linguistic perspective. Section 2 discusses existing analyses of expletive negation under ‘before’ cross-linguistically, proposals for the semantics of ‘before’, and also Lin 2016’s idea that *yiqian* is sensitive to a transition of state

³Under the categorization of Ogihara & Steinert-Threlkeld 2024, the three readings stand for three different subtypes of ‘before’ clauses. The first type is considered veridical ‘before’-clauses; the second type is called non-committal ‘before’-clauses; and the third type is called anti-veridical ‘before’-clauses.

of affairs, which lays out the conceptual foundation of the formal analysis of the current paper. Section 3 constitutes the core of our formal analysis and its empirical predictions, as we propose a formal explication of the *transition* operator and examine its interactions with different configurations involving Aktionsart, grammatical aspect, etc. Section 4 discusses the asymmetry between *yiqian* ‘before’ and *yihou* ‘after’ in terms of their compatibility with (semantically vacuous) negation, arguing that *yihou* selects for the *earliest* operator instead of the *transition* operator. Section 5 concludes the paper and offers some methodological discussions on approaches to expletive negation from a cross-linguistic perspective.

2 Background

2.1 Expletive Negation under ‘Before’ Cross-linguistically

The proposal of Espinal 2000 for Catalan and the proposal of Del Prete 2008 for Italian are partially motivated by the observations that in these two languages, expletive negation under *abans* and *prima*, respectively, is only licensed in non-veridical *abans/prima*-clauses. The following examples illustrate this constraint:

- (9) Context: Joan bought the car.
- a. En Joan revisà el cotxe abans de comprar lo.
Det Joan checked the car before of buying it_{CL}
‘Joan checked the car before he bought it.’
 - b. *En Joan revisà el cotxe abans que no el comprés.
Det Joan checked the car before that not it_{CL} bought.SUBJ.3SG
‘Joan checked the car before he bought it.’ (Catalan, Espinal 2000)
- (10) Saluterai Leo prima che (?non) parta per Roma.
Greet.2SG.FUT Leo before that not leave.3SG.SUBJ for Roma
‘You will see Leo off before he leaves for Rome.’ (Italian, Del Prete 2008)

The contrast between the two sentences in (9) suggests that in a context where the *abans*-clause in Catalan is veridical, expletive negation is not allowed under *abans*. Similarly, (10) suggests that the insertion of an expletive negation under Italian *prima* is deviant when the sentence is uttered in a context where Leo’s departure is taken for granted.

Further, expletive negation in Catalan and Italian occurs in other non-veridical environments besides under *abans* and *prima*, according to Espinal 2000. For instance, certain lexical items with non-veridicality property like ‘until’ and inequality comparative operators also license expletive negation in these two languages and some other Romance languages like Spanish.⁴ In other words, expletive negation in these languages is not induced by ‘before’ directly, but by the non-veridicality property carried by certain lexical items, including ‘before’. The examples below illustrate expletive negation licensed by *finché* ‘until’ in Italian, *hasta* ‘until’ in Spanish and the inequality comparative in Catalan:

⁴Korean interesting shows a similar empirical picture: ‘before’ and ‘until’ in Korean also license expletive negation. The former is reported in Jin & Koenig 2020, and the author is grateful to the Korean native speakers in Sinn und Bedeutung 30 for the latter.

- (11) Resto finché (non) arriva qualcuno.
 stay.1SG until not arrives somebody
 ‘I’ll stay until somebody arrives.’ (Italian, Espinal 2000)
- (12) Me quedaré hasta que (no) me echen.
 me_{CL} stay.FUT.1SG until that not me_{CL} throw.3PL
 ‘I will stay until they throw me out.’ (Spanish, Espinal 2000)
- (13) Valmés que vingueu que no que us quedeu sols.
 better that come.SUBJ.2PL than not that you_{CL} remain.2PL alone
 ‘It’s better you come than you stay on your own.’ (Catalan, Espinal 2000)

Motivated by these observations, Espinal 2000 (followed by Del Prete 2008) postulates a non-veridical operator in the syntactic structure when a negation occurs under a lexical item with non-veridicality property. The operator attracts the embedded negation via feature checking, and then negation absorption takes place, where the non-veridical operator eliminates the superfluous negative feature carried by the embedded negation, making it expletive.

The proposal cannot be imported into Mandarin, because Mandarin expletive negation under *yiqian* is not restricted to non-veridical environments, so we do not have the motivation for positing a non-veridical operator. Further, negation absorption proposed by Espinal 2000 over-generates in Mandarin, predicting that negation under lexical items with a non-veridicality property, such as *zhídào* ‘until’ and *fǒurèn* ‘deny’, would also be semantically vacuous, contra empirical facts.

On the other hand, Krifka’s 2012 semantic-pragmatic approach for German (see the appendix of Lin 2016 for a concise summary of this approach) crucially rely on the fact that German expletive negation under *bevor* is only licensed by a matrix negation. Expletive negation under *yiqian* in Mandarin, however, does not require a matrix negation, as we have seen in the previous section. Therefore, Krifka’s account for German cannot be extended to analyze the Mandarin phenomenon, either.

Since existing analyses for expletive negation under ‘before’ in other languages cannot account for the data pattern in Mandarin, we shall aim to propose a separate analysis for expletive negation under *yiqian*.

2.2 Previous Analyses for ‘Before’

We first examine previous analyses for ‘before’ to see whether they account for the data pattern shown by Mandarin *yiqian* clauses. One canonical analysis of ‘before’-clauses in English is proposed by Beaver & Condoravdi 2003 and echoed in Condoravdi 2010 (see some comments in Krifka 2010). Under this analysis, the complement proposition of ‘before’ combines with a type-coercion operator *earliest*, which selects the earliest time point (a definite time description) in the time interval denoted by the complement proposition. The function of ‘before’, then, is to specify the temporal order between this definite time description and the interval denoted by the matrix proposition - that is, a construction in the form of ‘*p* before *q*’ has the following logical form:

- (14) [*p* [before [[*earliest*] *q*]]]

The denotation of the *earliest* operator spelt-out below picks out the earliest time point of a time interval denoted by a proposition:

$$(15) \quad \llbracket \textit{earliest} \rrbracket = \lambda P_{<i,t>}. \lambda t. [\exists I [t \in I \wedge P(I) \wedge \forall t' [t' \in I \rightarrow t \leq t']]]$$

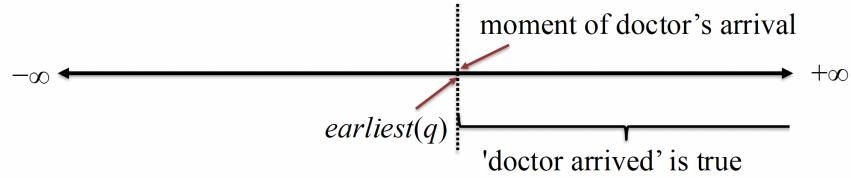
Now, a question naturally follows: does the *earliest* operator also co-occur with Mandarin *yiqian*? We answer this question by applying the *earliest* operator to our core data points (1-a) and (1-b), repeated below, to see whether we are able to yield the correct truth conditions.

- (16) a. yisheng lai yiqian, bingren jiu yijing si le.
 doctor come before patient then already die PFV.
 ‘Before the doctor arrived, the patient had already died.’
 b. yisheng **mei** lai yiqian, bingren jiu yijing si le.
 doctor NEG come before patient then already die PFV.
 ‘Before the doctor arrived, the patient had already died.’ (Adapted from Lin 2016)

Let’s first consider the positive variant. The intuitive understanding of Mandarin *yiqian*-clauses corresponds to a Perfect (Iatridou et al. 2003) interpretation, with the positive variant picking out the set of time points following the event time:

$$(17) \quad \llbracket \textit{yisheng lai} \rrbracket = \lambda t. [\exists e [\tau(e) \preceq t \wedge \textit{arrive}(e, \textit{doctor})]].$$

Applying the *earliest* operator to the set of event time above, we yield the earliest time point of the interval of ‘doctor arrived’ being true, which coincides with the moment of the doctor’s arrival. *Yiqian* then establishes the temporal sequence of this earliest time point and the time of the patient dying, deriving the desired truth conditions, as illustrated by the graph below, where *q* represents the proposition embedded under *Yiqian*:



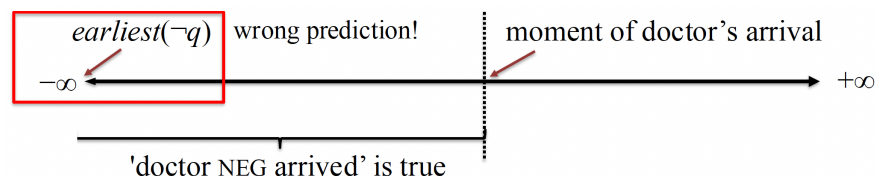
Graph 1: Applying the Earliest Operator to Positive Proposition under *Yiqian*

Let’s then consider the negative variant. The event time of the embedded proposition should be the complement of what we have for the positive variant on the time axis:

$$(18) \quad \llbracket \textit{yisheng NEG lai} \rrbracket = \lambda t. [\neg \exists e [\tau(e) \preceq t \wedge \textit{arrive}(e, \textit{doctor})]].$$

Applying the *earliest* operator to the set of event time above (which is a right-bounded set of time points, holding of all time points prior to the doctor’s arrival): $(-\infty, \tau(e)]$, it follows that the *earliest* operator picks out $-\infty$ when applied to this negative variant, which yields implausible truth conditions requiring the patient to have died before $-\infty$.⁵

⁵Applying the MAX operator adopted in Rett 2020 (following Rullmann 1995) yields the same result, because the MAX operator is assumed to pick the earliest time point as the point of maximal informativity when co-occurring with ‘before’.



Graph 2: Applying the Earliest Operator to Negative Proposition under *Yiqian*

In fact, Beaver & Condoravdi 2003 assume that *earliest* is only defined for left-bounded sets of time points, which correctly rules out English ‘before’-clauses with an embedded negation:

- (19) #Before John hasn’t entered college, ...

To capture data points with an embedded negation under *yiqian*, we will need a type-coercion operator that is similar to the *earliest* operator but can take either left-bounded or right-bounded sets of time points and return the correct boundary time point.

2.3 Lin 2016: Conceptual Basis

Lin 2016 argues that *yiqian* in Mandarin is not sensitive to the *earliest* operator like English ‘before’ is, but it is sensitive to a transition of state of affairs. Empirical evidence for this argument includes the observation that *yiqian* seems to select only complement clauses that involve a change of states; those propositions that are constantly true or false throughout the time axis and thus lack a transition of states cannot be embedded under *yiqian*.⁶

- (20) #Yi jia yi dengyu er yiqian, ...
 one plus one equal two before
 ‘Before one plus one equals two, ...’
- (21) #suoyoude sanjiaoxing dou you san-ge bian yiqian, ...
 all triangles all have three-CL side before, ...
 ‘Before all triangles have three sides, ...’

Following this observation, Lin proposes the following presupposition as part of the meaning of sentences in the form of ‘*P yiqian, Q*’:

- (22) ‘*P yiqian, Q*’ presupposes that a change of state associated with *P* from the prior non-existent state (PS) to the resultant post-state (RS) must be possible.

Crucially, the proposal above entails a transition time at which the change from PS to RS occurs; and a *transition* operator, parallel to the *earliest* operator in English, picks out this transition time. The function of *yiqian* is to specify the temporal order between this transition time and the interval denoted by the matrix proposition.

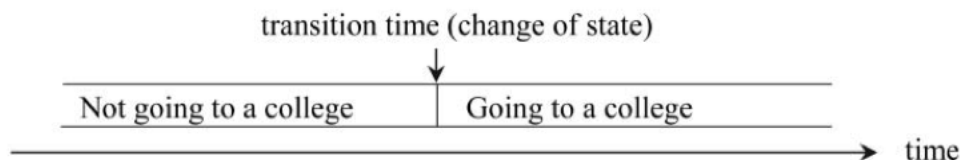
Further, it is assumed that an affirmative proposition typically expresses a resultant state, while a negative proposition typically expresses a prior state. Conceptually, the transition point coin-

⁶It is worth noting that the English translations in (12) and (13) are equally bad, and this is predicted by the nature of the *earliest* operator. The crucial contrast between English ‘before’ and Mandarin *yiqian* in fact lies in the ability to embed right-bounded sets of time points, which motivates our formal explication of the *transition* operator in later sections.

cides with the earliest moment of the resultant state and the latest moment of the prior state. For example, the affirmative proposition *yisheng lai* ‘doctor arrive’ is the resultant state and the negative proposition *yisheng mei lai* ‘doctor NEG arrive’ is the prior state. The transition point is the moment at which the doctor arrives: before this moment, the negative proposition holds true; after this moment the affirmative proposition holds true.

Consequently, when we have an affirmative proposition as the complement of *yiqian*, the transition operator would pick out the earliest moment in the interval denoted by the proposition; when we have the negative counterpart of the same proposition, the transition operator would pick out the latest moment in the interval denoted by the negative proposition. Crucially, these two moments are indistinguishable. In this way, embedding an affirmative proposition under *yiqian* yields the same result as embedding its negative counterpart. The embedded negation appears to be semantically vacuous but is in fact interpreted. Having the same truth conditions with or without the negation is due to the transition operator that co-occurs with Mandarin *yiqian*.

Lin 2016 uses a diagram to illustrate the idea sketched above, with the following sentence as example. The diagram illustrates the time intervals denoted by the affirmative version and the negative version of the proposition under *yiqian*:⁷



Graph 3: Lin’s Illustration of Transition of States

- (23) Zhangsan (mei) shang daxue yiqian, ta jiu yijing hen dongshi le.
 Zhangsan NEG attend university before he then already very sensible PAR
 Zhangsan had already been sensible before he went to a college.

As we can see in the diagram, the earliest time of ‘having ever gone to a college’ and the latest time of ‘not having ever gone to a college’ are indistinguishable; they coincide with the transition time, which is picked out by the transition operator. *Yiqian*, then, specifies the temporal order of the transition time and the time interval at which ‘Zhangsan had been sensible’ is true.

Lin 2016 provides a conceptual explanation for the puzzle of negation under *yiqian* in Mandarin, but we still do not have a spelt-out denotation for the *transition* operator that is parallel to the *earliest* operator proposed for English in Beaver & Condoravdi 2003 and Condoravdi 2010. Relatedly, it is not clear how the transition operator is able to distinguish between prior states and resultant states and select the correct time point. Section 3 of the current study aims to build on Lin’s proposal and discuss some empirical predictions that follow.

⁷Although not stated explicitly in Lin 2016, the time intervals representing the positive and negative variants of the *yiqian*-clause in the diagram captures the nuance that *yiqian*-clauses receives a Perfect interpretation: for the positive variant, the corresponding time interval is the set of all time points after the moment when Zhangsan entered college, and the complement of this time interval on the time axis corresponds to the negative variant. We will discuss more about how negation interacts with states and events in later sections.

3 The Core Analysis: the Transition Operator

This section advances a formal semantic analysis of Mandarin *yiqian*-clauses, arguing that the apparent vacuity of embedded negation is a predictable consequence of the composition of aspectual operators. Building on the conceptual basis in Lin 2016 that *yiqian* is sensitive to state transitions, I propose a formal explication of a novel *transition* operator ($\mathcal{T}$), which selects the unique non-infinite boundary of a time interval. I demonstrate that $\mathcal{T}$ operates within a two-level aspectual coercion system: an initial event-level coercion determined by Aktionsart (following Rett 2020), followed by the clause-level application of $\mathcal{T}$. Crucially, I argue that the interaction between these operators and negation is governed by the ontological distinction between states and events, necessitating a scopal distinction where negation applies low for stative predicates but high for non-stative predicates. This system not only resolves the initial puzzle of "expletive" negation in achievements (besides activities and accomplishments) but also correctly derives the non-vacuous interpretation of negation with stative predicates and the infelicity of embedded imperfectives under *yiqian*.

3.1 Formalizing the Transition Operator

Recall our discussion of the *earliest* operator proposed for English 'before' in Beaver & Condoravdi 2003 and Condoravdi 2010: this operator fails to yield the correct truth conditions when applied to *yiqian*-clauses with an embedded negation, e.g., *yisheng mei lai* 'the doctor did not arrive'. The semantic denotation of the negated *yiqian*-clause is the complement of the affirmative set, resulting in a right-bounded time interval $(-\infty, t_1]$. When the earliest operator applies to this set, it must select the earliest time point. Since the interval extends infinitely into the past, the operator either returns $-\infty$ or is undefined if we restrict the argument of the operator to left-bounded intervals, incorrectly predicting that the sentence means "before the beginning of time" or is ungrammatical. Thus, the operator responsible for temporal ordering in Mandarin *yiqian* constructions cannot be *earliest*.

Lin's 2016 idea suggests that the transition operator picks out the earliest time point of a resultant state, which corresponds to an affirmative proposition, and the latest time point of a prior state, which corresponds to a negative proposition. This treatment conceptually accounts for cases where achievements are embedded under *yitian*, as we have seen in previous examples, but it does not explain cases where the embedded proposition under *yiqian* is negative but a resultant state. Consider the following example, where a negated stative predicate is embedded under *yiqian*:

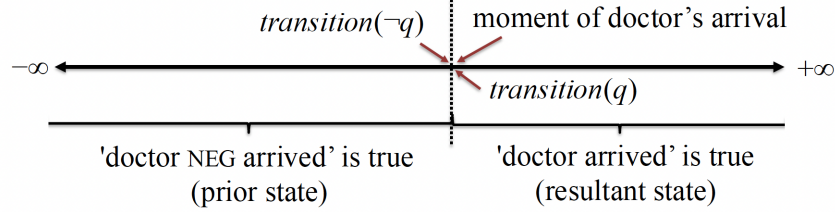
- (24) wo bu xihuan Zhangsan yiqian,...
 I NEG like Zhangsan before
 'Before I dislike Zhangsan, ...'

In this example, the *yiqian*-clause implies that there is a transition from neutral or liking to disliking. Therefore, the embedded proposition should be categorized as resultant state, and the transition operator should select the earliest time point of 'disliking'. However, following Lin's treatment, the negative proposition should be a prior state, and the *transition* operator selects the latest time point of 'disliking', which is an unattested reading.⁸

⁸One crucial difference between embedded states and other eventuality types is that negation is not vacuous in

To solve the predicament, we propose that the *transition* operator is agnostic about the state type of its complement proposition; rather, the operator solely checks the boundedness of its argument time interval. It picks out the earliest time point of a right-bounded time interval or the latest time point of a left-bounded time interval. For time intervals that are bounded on both ends or on neither ends, the *transition* operator returns undefined, as there is no unique time point that stands out from the time interval; or in the framework of Rett 2020, there is no maximally informative time point in these cases. Following this conceptual proposal, we formalize the *transition* operator:

$$(25) \quad \llbracket \mathcal{T} \rrbracket = \lambda Q_{<i,t>}. \iota t [t \in Q \wedge t \neq \pm\infty \wedge [\forall t' [t' \in Q \rightarrow t \leq t'] \vee \forall t'' [t'' \in Q \rightarrow t \geq t'']]].$$



Graph 4: Deriving Semantically Vacuous Negation with $\mathcal{T}$

The graph above illustrates how the *transition* operator $\mathcal{T}$ works conceptually. Unlike the earliest operator, which rigidly selects the left boundary, $\mathcal{T}$ is defined to select the unique non-infinite boundary of a time interval. For a left-bounded interval ($I_A = [t_1, +\infty)$), $\mathcal{T}$ picks out its earliest time point t_1 ; for a right-bounded interval ($I_B = (-\infty, t_2]$), $\mathcal{T}$ picks out its latest time point t_2 . This provides a formal explanation of the expletive nature of negation under Mandarin *yiqian*: for a left-bounded time interval embedded under *yiqian* like the (1-a) that denotes $Q_A = [t_1, +\infty)$, its starting point t_1 is effectively indistinguishable from the end point t_2 of its negative counterpart, which denotes an interval $Q_B = (-\infty, t_2]$. That is, t_1 and t_2 coincide with the transition point. Since t_1 and t_2 are effectively indistinguishable, the positive and negative variants of a *yiqian*-clause yield the same truth conditions, and therefore negation under *yiqian* appears semantically vacuous.

3.2 Empirical Support: Vendler's Four Classes

In this section, we integrate the *transition* operator $\mathcal{T}$ into a comprehensive compositional framework to account for a wide range of empirical data. Our analysis relies on two core formal ingredients. First, we posit a two-level system of aspectual coercion, where an initial event-level aspectual coercion (determined by Aktionsart, proposed in Dölling 2014 and echoed in Rett 2020 for her analysis of English 'before') precedes the application of the clause-level *transition* operator, which is assumed to be a kind of aspectual coercion as well. Second, we propose a scopal hypothesis regarding negation and stativity, arguing that the fundamental ontological distinction

negated embedded states. For example, the following sentence implies a transition from neutral or disliking to liking, the exact opposite of (24).

- (i) wo xihuan Zhangsan yiqian,...
I like Zhangsan before
'Before I like Zhangsan, ...'

In later sections, we discuss embedded states under *yiqian* in more details and show that our system correctly derives the non-vacuous interpretation of negation with stative predicates.

between states and events dictates the structural position of the negative operator relative to aspectual coercion. We then demonstrate how these ingredients interact to correctly derive the truth conditions for all four of Vendler’s aspectual classes, providing a unified account for why negation appears vacuous with eventive predicates (activities, accomplishments and achievements) but remains interpreted with stative predicates.

3.2.1 The Two-Level System for Aspectual Coercion

In Rett 2020, English ‘before’-clauses are argued to have two levels of aspectual coercion: one is achieved via the MAX operator, whose application result is directly feed to ‘before’ to establish the desired temporal sequence; the other is achieved by an operation on the bare eventuality level, coercing the event time into a single time point based on the type of the eventuality. We call the former ‘clause-level aspectual coercion’, since it is immediately below ‘before’ in the structure and takes the whole clause embedded under ‘before’ as its argument; besides the MAX operator, the *earliest* operator in Beaver & Condoravdi 2003 and our $\mathcal{T}$ operator are also assumed to facilitate clause-level aspectual coercion. On the other hand, we call the latter type ‘event-level aspectual coercion’, as it applies directly to the eventuality before other elements (e.g., tense) participate in the composition. Following Rett 2020, atelic eventuality (including states and activities) selects inchoative coercion, which coerces the event time interval into its starting point; telic eventuality (including accomplishments and achievements) selects completive coercion, which coerces the event time interval into its end point. Rett 2020 assumes that event-level coercion is optional in English ‘before’-clauses to account for the ambiguity involved in certain ‘before’-clauses. Consider the example below:

(26) John met Mary before she climbed the mountain.

The sentence above can mean either ‘John met Mary before she started to climb the mountain’ or ‘John met Mary before she summited mountain’. Under the analysis of Rett 2020, to derive the former meaning, completive coercion is not applied, so only the MAX operator (which selects the earliest time point of a time interval in ‘before’-clauses) facilitates clause-level coercion, and ‘before’ establishes the temporal sequence of the earliest time of mountain climbing and the time of meeting. In comparison, when completive coercion is indeed applied, the latest time of the eventuality will be fed into the MAX operator instead, which trivially returns the same time point, and ‘before’ therefore establishes the temporal sequence of the latest time of mountain climbing (i.e., the time point when Mary summited the mountain) and the time of meeting. The following schema replicated from Rett 2020 illustrates the derivation of the two different readings, supposing John met Mary at 3pm and Mary climbed the mountain from 1pm to 4pm:

(27) John met Mary before she climbed the mountain.

- a. $= 3\text{pm} \prec \text{MAX}([1\text{pm}, 4\text{pm}]_{\prec})$
 $= 3\text{pm} \prec 1\text{pm}$ $\prec$ initial: false
- b. $= 3\text{pm} \prec \text{MAX}(\text{completive}([1\text{pm}, 4\text{pm}]_{\prec}))$
 $= 3\text{pm} \prec \text{MAX}([4\text{pm}]_{\prec})$
 $= 3\text{pm} \prec 4\text{pm}$ $\prec$ final: true

It is worth noting that the optionality of event-level coercion is not an inherent property of the oper-

ation itself, but rather varies across languages. In the original proposal of Dölling 2014, an abstract, underspecified coercion operator (coerce*) is mandatorily inserted into the semantic structure of all event predicates, which is instantiated by pragmatics. Some languages encode overt aspectual markers to achieve the event-level coercion when it is required. Serbo-Croatian is one such language (examples from Rett 2020):

- (28) a. Mary je sreća Johna pre nego što se
 Mary-NOM is-PRES-3FS met-PP-3FS John-ACC before than PTCL REFL
 peo na vrh planine.
 climb-IMP-3MS on top-ACC mountain-GEN
 ‘Mary met John before he climbed to the top of the mountain.’
 b. Mary je sreća Johna pre nego što se
 Mary-NOM is-PRES-3FS met-PP-3FS John-ACC before than PTCL REFL
 popeo na vrh planine.
 climb-PP-3MS on top-ACC mountain-GEN
 ‘Mary met John before he climbed to the top of the mountain.’

As we can see from the difference in the morphological markers on the verb ‘climb’ in the above two variants, whether completive coercion is involved depends on the presence and absence of an overt aspectual marker. On the other hand, some languages block aspectual coercion altogether. Tagalog is one example (examples from Rett 2020):

- (29) Nakilala ni-Mary si-John bago siya um-akyat sa bundok.
 PFV.TV-meet GEN-Mary SUBJ-John before SUBJ.3SG PFV.AV-climb OBL mountain
 ‘Mary met John before he climbed to the top of the mountain.’

The sentence above is unambiguous; only the ‘before initial’ reading is available.

Circling back to *yiqian*-clauses in Mandarin, we will assume that event-level aspectual coercion is always required under *yiqian*-clauses, perhaps under the pressure of the Perfect operator as a default component of *yiqian*-clauses, in order to derive the correct interpretations with different types of eventuality embedded under *yiqian*. We adapt the formal explications of inchoative coercion and completive coercion in Rett 2020 to accommodate the assumed obligatoriness. The following constitutes one of the formal ingredients for our composition of Mandarin *yiqian*-clauses and forms the two-level aspectual coercion system together with our $\mathcal{T}$ operator:

- (30) a. inchoative coercion: If e is a process with duration T , inchoatively coerced e denotes $\text{GLB}(T)$, where $\text{GLB}(T) = \iota t[t \in T \wedge \forall t' \in T[t \preceq t']]$.
 b. completive coercion: If e is a culmination with duration T , completively coerced e denotes $\text{LUB}(T)$, where $\text{LUB}(T) = \iota t[t \in T \wedge \forall t' \in T[t \succeq t']]$.

3.2.2 The Scopal Hypothesis: States vs. Non-states

Another formal ingredient of our analysis concerns the scope of negation relative to the event-level aspectual coercion operators defined in the previous subsection. We propose that the structural position of negation is non-uniform, determined by the fundamental ontological distinction between stative and eventive predicates. Intuitively, negating a stative predicate results in a condition that is itself state-like; for instance, the negation of the state ‘Zhangsan knowing the answer’ yields

‘Zhangsan not knowing the answer’, which holds as a stable, distinct state having its own temporal boundaries. In contrast, negating an eventive predicate, such as the achievement ‘the doctor arrived’ does not produce a new achievement event; rather, it simply asserts the absence (i.e., non-existence) of an event instantiation. Because negated states retain their stative properties, we assume that negation applies before inchoative coercion (i.e., negation scopes low), allowing the aspectual operator to target the onset of the ‘negative state’. On the other hand, when events (including activities, accomplishments and achievements) are negated, we assume that negation applies only after the event has been composed with Perfect to yield the set of time points after the coerced event time, and negation picks out a consecutive time interval that is the complement of this set of time points on the time axis. The following schema illustrates the crucial structural difference we have proposed:

- (31) a. $[yiqian [\mathcal{T} [Perfect [Inch. [NEG P]]]]]$, if P involves a stative predicate.
 b. $[yiqian [\mathcal{T} [NEG [Perfect [Inch./Comp. P]]]]]$, if P involves an eventive predicate.

One might object that postulating different scopes for negation for different eventuality types is an ad hoc solution; however, this distinction is strongly supported by the ontological findings in Kusumoto 2011. She supports the distinction between states and events with evidence from the Japanese temporal connective *aida-ni* ‘while’, which selects exclusively for states; crucially, *aida-ni* accepts negated states but rejects negated events, proving they are semantically distinct objects. Further, she proposes formal denotations for negation as well as stative and eventive predicates while further fixing a sophisticated denotation for Japanese *aida-ni* ‘while’ with sub-state/event relations to account for the distribution of different eventuality types under *aida-ni*. We adopt her treatment of stative and eventive predicates:

- (32) a. Denotation of stative predicates, using ‘be in the room’ as an example:
 $\lambda s \in D_{ev}. [\lambda x \in D_e. be - in - the - room(s)(x)]$
 b. Denotation of eventive predicates, using ‘run’ as an example:
 $\lambda e \in D_{ev}. [\lambda x \in D_e. run(e)(x)]$

This is in fact a unified account for stative predicates and eventive predicates. We will need further mechanisms to derive the difference between negated states and negated events: we achieve this by postulating our scopal hypothesis for negation; Kusumoto 2011 achieves this with a complicated semantics of the temporal connective *aida-ni* ‘while’. Therefore, we deviate from her treatment of negation and assume the following formal implementation:

- (33) a. Stative predicates and eventive predicates undergo existential closure and are composed with the subject, forming the embedded proposition P of type $\langle i, t \rangle$:
 $\llbracket P \rrbracket = \lambda t. \exists e [Q(e) \wedge t \in \tau(e)]$, where Q is the property (subject and predicate) of the eventuality and $\tau(e)$ is the time interval when the eventuality holds.
 b. One unified denotation of negation, which simply takes a time interval as its argument and returns the complement of this time interval on the time axis:
 $\llbracket NEG \rrbracket = \lambda I. \lambda t. t \notin I$
 c. A simple treatment of *yiqian* following the canonical proposal for ‘before’, which establishes the temporal sequence of a time point specified by the embedded clause and the matrix time interval:

$$\llbracket yiqian \rrbracket = \lambda t. \lambda I. I \prec t$$

We have now gathered all the formal ingredients we need to compositionally derive the semantics of Mandarin *yiqian*-clauses: the *transition* operator $\mathcal{T}$, the two types of event-level aspectual coercion in (30), our structural assumptions in (31), our denotations for embedded positive propositions under *yiqian* and negation in (33), as well as a semantics for *yiqian*.⁹

3.2.3 The Empirical Payoff: the Transition Operator and Eventuality Types

In this section, we consider Vendler’s four classes of predicates: state, activity, accomplishment, and achievement and show that our proposed formal mechanisms capture the empirical patterns of these types of predicates when embedded under *yiqian* ‘before’.

For embedded states under *yiqian*, we make the following empirical generalizations: embedded states are forced to be resultant states, i.e., states that follow an implicit transition point (and thus are left-bounded set of time points); embedding a state under *yiqian* means ‘before the initial moment of the state’. The former generalization is illustrated by the following patterns:

- (34) a. Zhangsan xihuan baicai yiqian, conglai bu chi shucai.
 Zhangsan like cabbage before ever NEG eat vegetable
 ‘Before Zhangsan liked cabbage, he never ate any vegetable.’
 b. Zhangsan bu xihuan baicai yiqian, shenti feichang jiankang.
 Zhangsan NEG like cabbage before body very healthy
 ‘Before Zhangsan disliked cabbage, he was very healthy.’

The positive (a) variant above implies that there exists a transition from neutral or disliking to liking, and the state embedded under *yiqian* is therefore a resultant state. The negated (b) variant above implies that there exists a transition from neutral or liking to disliking, and therefore the state embedded under *yiqian* in this variant is still a resultant state. Embedding a prior state is infelicitous, as we can see from the following examples, assuming that the transition from ‘not knowing’ to ‘knowing’ is natural, but the reverse is highly unusual:

⁹The semantics we adopt for *yiqian* is a simplifying assumption without any consideration of intensionality. With more complicated configurations, we will need *yiqian* ‘before’ to establish more nuanced temporal sequences. For example, when we have anti-veridical *yiqian*-clauses:

- (i) Mozhat (mei) wancheng anhunqu yiqian, jiu si le.
 Mozart NEG finish Requiem before then die PFV
 ‘Mozart died before he finished the Requiem.’ (Adapted from Ogiwara & Steinert-Threlkeld 2024)
 (ii) zhadan (mei) baozha yiqian, jiu bei chaichu le.
 bomb NEG explode before then PASS remove PFV
 ‘Bombs were defused before they exploded.’ (Adapted from Lin 2016)

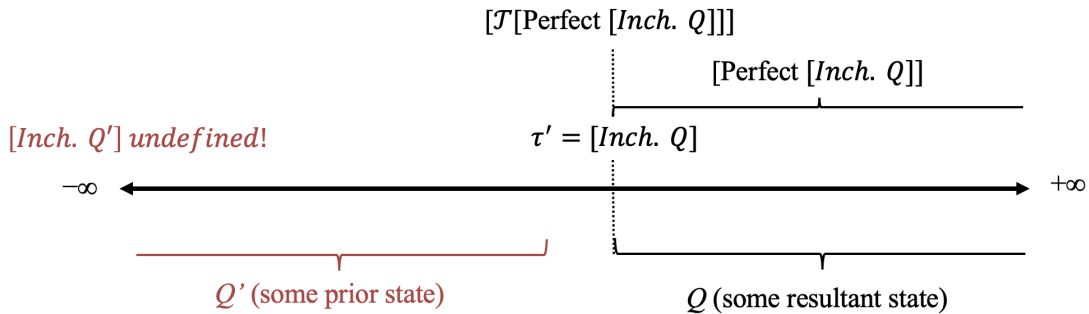
In the two examples above, the *yiqian*-clauses cannot be true in the actual world and are thus considered anti-veridical. Nevertheless, we can establish the temporal sequence between the embedded propositions and the matrix propositions, which were realized in the actual world. Ogiwara & Steinert-Threlkeld 2024 proposed a syncategorematic solution involving a system of relevant alternative worlds to characterize the semantics of anti-veridical ‘before’-clauses. For space reasons, we do not import their system here. How our $\mathcal{T}$ operator works with anti-veridical *yiqian*-clauses is an interesting topic for future research.

- (35) a. Zhangsan zhidao da'an yiqian, feichang jiaolv.
 Zhangsan know answer before very anxious
 'Before he knew the answer, Zhangsan was very anxious.'
 b. #Zhangsan bu zhidao da'an yiqian, ...
 Zhangsan NEG know answer before
 '#Before Zhangsan didn't know the answer...'

The (a) variant above is coherent, since the implication of a transition point from 'not knowing' to 'knowing' is easily accepted. If we keep this implication and force the embedded state under *yiqian* to be a prior state, as we have in the (b) variant above, the sentence becomes infelicitous. The only way to coerce a meaning out of the (b) variant above is to assume a context involving amnesia to justify a transition from 'knowing' to 'not knowing', which in turn forces the embedded state into a resultant state. One further implication of the 'resultant state only' restriction is that states lacking a transition point (always-true states) cannot be embedded under *yiqian* (also serves as part of the empirical motivation for Lin's 2016 proposal):

- (36) #Sanjiaoxing dou you san tiao bian yiqian, ...
 triangle all have three CL side before
 '#Before all triangles have three sides, ...' (Adapted from Lin 2016)

Our proposed formal mechanisms successfully capture the data patterns above. Recall that we assume event-level aspectual coercion is obligatory in Mandarin *yiqian*-clauses under the pressure of *Perfect*, and for events, the available event-level aspectual coercion is inchoative coercion in (30-a), which coerces a time interval into its earliest time point. This is problematic for prior states, which are right-bounded set of time points spanning from $-\infty$ to the implicit transition point: applying inchoative coercion to embedded prior states will infinitely backtrack, since there is no unique time point such that it is the earliest time point of the event time interval. The same reasoning apply to always-true states, which are unbounded sets of time points that lack an earliest time point. Prior states and always-true states are thus ruled out under *yiqian* following our formal explications. Further, our proposed mechanisms correctly predict that *yiqian* points to the initial time point of an embedded resultant state, illustrated by the following graph and derivation:



Graph 5: Deriving the Interpretation of Embedded States under *Yiqian*

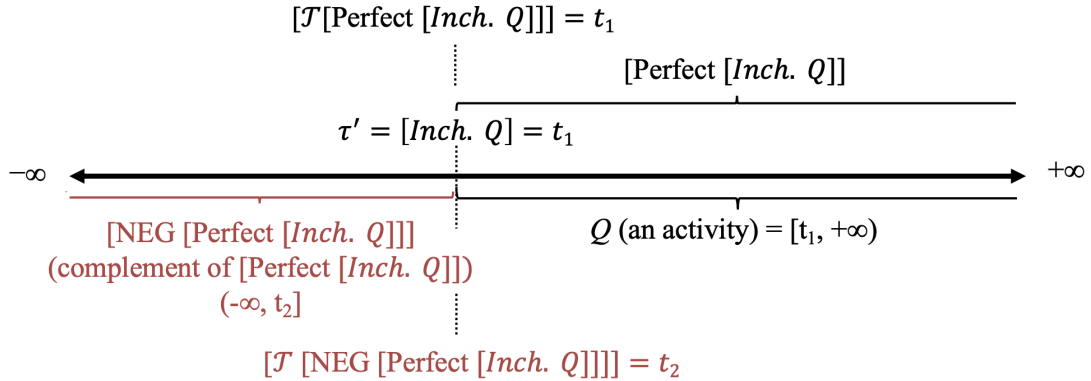
- (37) a. The embedded state undergoes inchoative coercion and yields the coerced event time τ' :
 $\tau' = \llbracket [Inch. Q] \rrbracket = it. \exists e [Q(e) \wedge t \in \tau(e) \wedge \forall t' \in \tau(e) [t \preceq t']]$

- b. *Perfect* selects the set of time points that follow the coerced event time τ' :
 $\llbracket [Perfect [Inch. Q]] \rrbracket = \lambda t''. \tau' \preceq t''$
- c. Our $\mathcal{T}$ operator selects the one unique boundary of the time interval from the previous step, which is the initial time point τ'' (the moment right after τ' , indistinguishable from τ') as desired:
 $\llbracket [\mathcal{T} [Perfect [Inch. Q]]] \rrbracket = \tau''$
- d. Finally *yiqian* establishes the temporal sequence of τ'' and the matrix time interval, yielding the correct truth conditions.

Next, we consider embedded activities under *yiqian*. The empirical generalization is that negation seems semantically vacuous with embedded activities under *yiqian*, and the crucial time point involved in the temporal sequence established by *yiqian* is the initial moment of the eventuality in the positive variant. Notice the following example:

- (38) Zhangsan zuotian (mei) youyong yiqian, chi le yi kuai dangao.
 Zhangsan yesterday NEG swim before eat PFV one CL cake
 ‘Before Zhangsan swam yesterday, he ate a piece of cake.’

Both the positive and negative variants of the *yiqian* clause point to the time point when Zhangsan started swimming, and therefore the negation seems vacuous. Our proposed formal mechanisms capture this pattern. Recall our scopal hypothesis for negation: with eventive predicates, negation applies to the result of applying *Perfect* and returns its complement time interval on the time axis. Based on this hypothesis, the derivation is illustrated by the following graph and steps:



Graph 6: Deriving the Interpretation of Embedded Activities under *Yiqian*

- (39) a. The embedded activity, which is atelic and durative, undergoes inchoative coercion and yields the coerced event time τ' , which is the initial moment of the activity:
 $\tau' = \llbracket [Inch. Q] \rrbracket = \iota t. \exists e [Q(e) \wedge t \in \tau(e) \wedge \forall t' \in \tau(e) [t \preceq t']]$ ¹⁰
- b. *Perfect* selects the set of time points that follow the coerced event time τ' :
 $\llbracket [Perfect [Inch. Q]] \rrbracket = \lambda t''. \tau' \preceq t''$

¹⁰Note that Q here can be either positive or negated, since based on our scopal hypothesis for negation, states are negated before undergoing inchoative coercion; this crucially derives the difference between negated states and negated events.

- c. For the negative variant, the negation applies to the result of the previous step, yielding its complement on the time axis:

$$\llbracket \text{NEG} [\text{Perfect} [\text{Inch. } Q]] \rrbracket = \lambda t''' . t''' \preceq \tau'$$
- d. When $\mathcal{T}$ operator applies to the positive variant, it selects the one unique boundary of the time interval, which is the initial time point t_1 in the graph, coinciding with τ' , as desired:

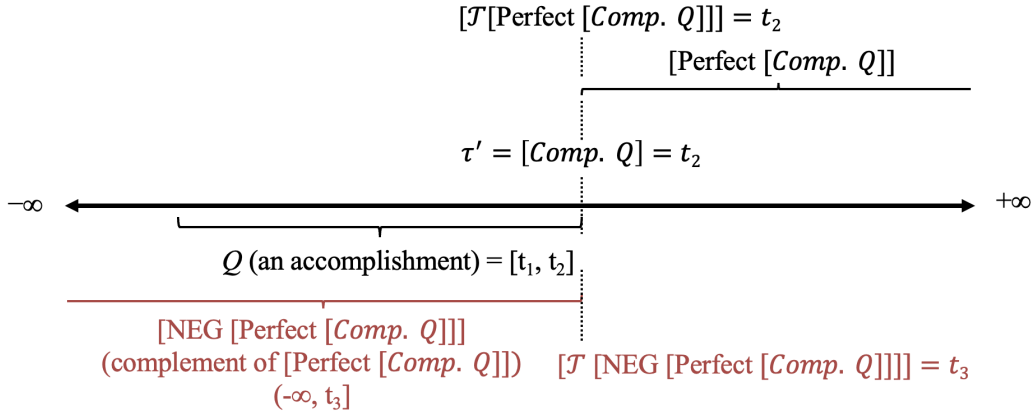
$$\llbracket [\mathcal{T} [\text{Perfect} [\text{Inch. } Q]]] \rrbracket = t_1$$
- e. When $\mathcal{T}$ operator applies to the negative variant, it selects the one unique boundary of the time interval, which is the final time point t_2 in the graph, coinciding with τ' , as desired:

$$\llbracket [\mathcal{T} [\text{NEG} [\text{Perfect} [\text{Inch. } Q]]]] \rrbracket = t_2$$
- f. Since t_1 and t_2 both coincide with τ' , they are indistinguishable, rendering the negation seemingly vacuous.
- g. Finally, *yiqian* establishes the temporal sequence of t_1/t_2 and the matrix time interval, yielding the same correct truth conditions.

We now consider embedded accomplishments under *yiqian*. Similarly, negation seems semantically vacuous with embedded accomplishments under *yiqian*, but the crucial time point is the final, culmination moment of the eventuality (recall that accomplishments are telic and durative), as exemplified by the following sentence:

- (40) Zhangsan (mei) xie shi feng xin yiqian, bu hui qu kan dianying.
 Zhangsan NEG write ten CL letter before NEG HUI go see movie
 'Before Zhangsan wrote ten letters, he wouldn't go to see a movie.'

Both the positive and negative variants point to the time point when Zhangsan finish(ed) writing ten letters, and the negation therefore seems vacuous. We now illustrate how our proposed system accounts for this pattern with the following graph and derivation steps:



Graph 7: Deriving the Interpretation of Accomplishments under *Yiqian*

- (41) a. The embedded accomplishment, which is telic and durative, undergoes complete coercion and yields the coerced event time τ' , which is the final, culmination moment of the accomplishment:

$$\tau' = \llbracket [\text{Comp. } Q] \rrbracket = \iota t . \exists e [Q(e) \wedge t \in \tau(e) \wedge \forall t' \in \tau(e) [t \succeq t']]$$

- b. *Perfect* selects the set of time points that follow the coerced event time τ' :
 $\llbracket [Perfect [Comp. Q]] \rrbracket = \lambda t'' . \tau' \preceq t''$
- c. For the negative variant, the negation applies to the result of the previous step, yielding its complement on the time axis:
 $\llbracket [NEG [Perfect [Comp. Q]]] \rrbracket = \lambda t''' . t''' \preceq \tau'$
- d. When $\mathcal{T}$ operator applies to the positive variant, it selects the one unique boundary of the time interval, which is the initial time point t_2 in the graph, coinciding with τ' , as desired:
 $\llbracket [\mathcal{T} [Perfect [Comp. Q]]] \rrbracket = t_2$
- e. When $\mathcal{T}$ operator applies to the negative variant, it selects the one unique boundary of the time interval, which is the final time point t_3 in the graph, coinciding with τ' , as desired:
 $\llbracket [\mathcal{T} [NEG [Perfect [Comp. Q]]]] \rrbracket = t_3$
- f. Since t_2 and t_3 both coincide with τ' , they are indistinguishable, rendering the negation seemingly vacuous.
- g. Finally, *yiqian* establishes the temporal sequence of t_2/t_3 and the matrix time interval, yielding the same correct truth conditions.

Our initial discussion of ‘before the doctor arrived’ falls into the category of achievements, which we turn to now with more detailed discussions. Attentive readers have noticed that in Section 3.1, we derived the semantic vacuity of the negation in embedded achievements without introducing event-level aspectual coercion. This does not indicate that our introduction of event-level aspectual coercion is ad-hoc and superfluous in some cases. Rather, it is due to the very nature of achievements: they are non-durative, and completive coercion therefore applies vacuously. The derivation of embedded achievements under *yiqian* in our proposed formal system is similar to that of embedded accomplishments, except for the fact that the coerced event time τ' yielded by completive coercion is identical to the one moment that the eventuality holds.

We have now shown the empirical evidence from different eventuality types embedded under *yiqian* for our proposed formal explication.

3.3 Further Support: Predicted Ungrammaticality

So far, we have considered cases where *yiqian*-clauses have their default interpretations with *Perfect*. What would our $\mathcal{T}$ operator predict with respect to *yiqian*-clauses that are not interpreted as having *Perfect*, for example, those *yiqian*-clauses with overt grammatical aspect markers? Our initial observation suggests that embedding imperfectives under *yiqian* results in infelicity, as we can see from the following example with the overt imperfective marker *zhe*:

- (42) Zhangsan xie (#zhe) zuoye yiqian, ...
 Z write IMPF homework before
 ‘Before Z is/was doing his homework, ...’

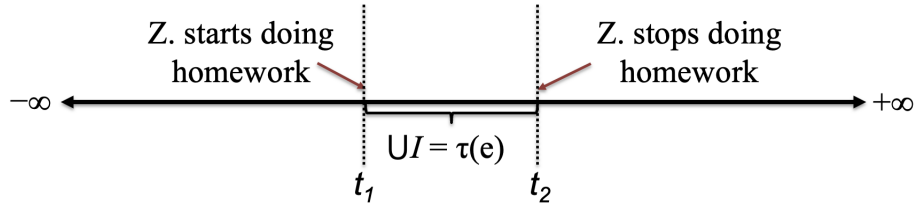
We argue that our proposed $\mathcal{T}$ operator captures the infelicity. Since we no longer have *Perfect* in our structure, we assume that event-level aspectual coercion lacks motivation and proposes the following simple structure for the *yiqian*-clause above:

$$(43) \quad [\text{yiqian} [\mathcal{T} [\text{Imperfective } Q]]]$$

Further, we adopt the standard treatment of imperfectives as in Kratzer 1998, using ‘Zhangsan’s doing homework’ as our embedded Q :

$$(44) \quad \llbracket [\text{Imperfective } Q] \rrbracket = \lambda I_{\langle i, t \rangle}. \exists e [\tau(e) \supseteq I \wedge \text{do.homework}(e, \text{Zhangsan})]$$

The above denotation amounts to the set of all time intervals that are contained in the period when Zhangsan does his homework. Following Beaver & Condoravdi 2003, we keep an ontological distinction between time points and time intervals but assume that consecutive time points can collapse into continuous time intervals. Therefore, the set denoted above can ultimately be collapsed into one single continuous interval, whose starting point t_1 is the moment when Zhangsan starts doing his homework, and the final point t_2 is the moment when Zhangsan finishes doing his homework. The graph below demonstrates the temporal operations described:



Graph 8: Deriving the Infelicity of Imperfectives under *Yiqian*

For the next step of the derivation, we will need to apply our $\mathcal{T}$ operator to the time interval $[t_1, t_2]$, which is crucially bounded on both sides.¹¹ Therefore, applying $\mathcal{T}$ to this time interval returns undefined as the uniqueness requirement of the ι operator is not satisfied, accounting for the infelicity of embedding imperfectives under *yiqian*.

4 The Asymmetry with *Yihou* ‘After’

Previous approaches have attempted to analyze ‘before’ and ‘after’ as lexical converses that differ only in the directions of the temporal sequences they establish (see Anscombe 1964 for counterarguments). Beaver & Condoravdi 2003 posited the *earliest* operator for ‘before’ and the *latest* operator for ‘after’, making the two a mirror image of each other both in terms of their semantics and the operators they select. More recently, Rett 2020 proposed a unified analysis for the temporal operator selected by ‘before’ and ‘after’ with the MAX operator, further adding to the

¹¹The most straight forward time intervals that are bounded on both sides are expressed by duration terms. We observe interesting speaker variations with respect to embedded duration terms under *yiqian*, for example:

- (i) %wuyue dao liuyue yiqian, ...
 May to June before
 ‘Before the period of time from May to June, ...’

Some speakers deem such phrases infelicitous as the $\mathcal{T}$ operator would predict, while some speakers have the ‘initial point’ interpretation as if the earliest operator applies (i.e., ‘June’ does not provide any useful information in the example above). It might be the case that *yiqian* works essentially different with nominal complements like duration terms. We leave this front of research to future investigations.

common properties of the two temporal connectives. Although symmetric analyses for ‘before’ and ‘after’ are elegant and attractive, there are striking differences between ‘before’-clauses and ‘after’-clauses when we adopt a cross-linguistic perspective and consider more configurations. For example, in Mandarin temporal clauses introduced by *yihou* ‘after’, we never observe the same type of semantically vacuous negation as in ‘*yiqian*’-clauses. In fact, embedding negation under *yihou* ‘after’ results in infelicity unless the embedded predicate is stative; in cases that do allow embedded negation, the interpretation of negation is never vacuous:

- (45) a. Zhangsan bu xihuan baicai yihou, zaiye mei chi guo baicai.
 Zhangsan NEG like cabbage after again NEG eat PFV cabbage
 ‘After Zhangsan didn’t like cabbage, he hadn’t eaten cabbage ever.’¹²
 b. Zhangsan xihuan baicai yihou, (??zaiye mei chi guo baicai.)
 Zhangsan like cabbage after again NEG eat PFV cabbage
 ‘After Zhangsan liked cabbage, he hadn’t eaten cabbage ever.’

This patterns with embedded states under *yiqian*. The contrast between the coherence of following up with the same sentence after the positive and negative variants of the *yihou*-clause indicates that the interpretation of the negation is by no means vacuous. However, the infelicity of embedding negated activities, accomplishments and achievements manifests crucial asymmetries between *yiqian* and *yihou*.¹³

- (46) a. #Zhangsan mei youyong yihou, ...
 Zhangsan NEG swim after
 ‘#After Zhangsan didn’t swim, ...’
 b. #Zhangsan mei xie shi feng xin yihou, ...
 Zhangsan NEG write ten CL letter after
 ‘#After Zhangsan didn’t write ten letters, ...’
 c. #yisheng mei lai yihou, ...
 doctor NEG come after
 ‘#After the doctor didn’t arrive, ...’

Although Lin 2016 did not discuss the asymmetry between *yiqian* and *yihou* with respect to their interactions with negation, he did make the following empirical generalization about embedding positive propositions under *yihou*:

- (47) *...the reference time of yihou-clauses is determined by aspectual classes and is always the initial time, unless the complement clause of after is an accomplishment that picks out the latest time of the interval at which the complement clause is true as the reference time.*
 (Lin 2016)

We argue that there is no need to make an ad hoc exception for embedded accomplishments. The only assumption we make is that *yihou* exclusively selects the earliest operator while *yiqian* ex-

¹²The English translation is judged to be marked/degraded; a more natural paraphrase is ‘after Zhangsan started disliking cabbage, ...’. There are apparently more nuances with English ‘after’-clauses, which fall out of the scope of our current study.

¹³We use past tense in the English translation for the sentences below, but the original Mandarin examples are unspecified for past or non-past interpretations.

clusively selects the $\mathcal{T}$ operator. Based on this assumption, together with the two-level aspectual coercion system we adopted for *yiqian*-clauses and our scopal hypothesis for negation and eventuality types, we can derive the correct interpretation for embedded accomplishments under *yihou*:

- (48) a. *yihou* is the lexical converse of *yiqian*, thus their semantics only differs in the direction of the temporal sequence:

$$\llbracket yihou \rrbracket = \lambda t. \lambda I. I \succ t$$
- b. The embedded accomplishment, which is telic and durative, undergoes completive coercion and yields the coerced event time τ' , which is the final, culmination moment of the accomplishment:

$$\tau' = \llbracket [Comp. Q] \rrbracket = \iota t. \exists e [Q(e) \wedge t \in \tau(e) \wedge \forall t' \in \tau(e) [t \succeq t']]$$
- c. *Perfect* selects the set of time points that follow the coerced event time τ' :

$$\llbracket [Perfect [Comp. Q]] \rrbracket = \lambda t'' . \tau' \preceq t''$$
- d. When the *earliest* operator applies to the left-bounded set of time points above, it selects the one unique boundary of the time interval, which is the initial time point, coinciding with τ' (the final moment of the accomplishment), as desired:

$$\llbracket [earliest [Perfect [Comp. Q]]] \rrbracket = \tau'$$
- e. Finally, *yihou* establishes the temporal sequence of τ' and the matrix time interval, yielding the correct truth conditions.

If the embedded accomplishment is negated, then we will have to apply the *earliest* operator to the following set of time points:

(49)
$$\llbracket [NEG [Perfect [Comp. Q]]] \rrbracket = \lambda t''' . t''' \preceq \tau'$$

This application would be problematic because the time interval formed by this set of time points is bounded on its right side only, and applying the *earliest* operator returns undefined, accounting for the infelicity of embedded negative accomplishments. The correct interpretation of embedded positive and negative states as well as the infelicity of embedded negative activities and achievements can be derived with the formal mechanisms and assumptions in similar manners.

However, we still do not have an explanatory theory as of why *yiqian* selects for the $\mathcal{T}$ operator, but *yihou* selects for the *earliest* operator. Our current study can only offer some speculations on this question. One glaring contrast between *yiqian*-clauses and *yihou*-clauses lies in their veridicality settings (see early discussions in Heinämäki 1978 and Ogihara 1995). While *yiqian*-clauses can be veridical, non-committal or anti-veridical (the veridicality relies heavily on the context), *yihou*-clauses are uniformly veridical in some sense: with past tense, the proposition embedded under *yihou* is taken to be true in the actual world; with non-past tense, the embedded proposition must be true in the speaker's epistemic alternatives (i.e., the speaker must be certain of the future realization of the embedded proposition in order for the *yihou*-clause to be uttered felicitously). The striking contrast between *yiqian* and *yihou* in their veridicality properties might lead them to be sensitive to different types of temporal junctures and thus select different aspectual coercion operators. Future investigation can further examine the interaction of different temporal connectives and veridicality.

5 Conclusion

This paper has advanced a formal semantic analysis of Mandarin *yiqian*-clauses, arguing that the puzzling phenomenon of semantically vacuous negation embedded under *yiqian* is a predictable consequence of the composition of negation and aspectual operators. We demonstrated that previous analyses of English ‘before’, which rely on the *earliest* operator (Beaver & Condoravdi 2003) or the MAX operator (Rett 2020), cannot account for the Mandarin data, particularly the acceptability of embedding negated propositions that denote right-bounded time intervals.

To solve the puzzle, we built on Lin’s 2016 conceptual basis of the *transition* operator and proposed a formal explication of this operator $\mathcal{T}$, which targets the unique non-infinite boundary of a temporal set. We integrated this operator into a two-level aspectual coercion system (also adopted in Rett 2020), where an initial, obligatory event-level coercion (determined by Aktionsart) feeds into the clause-level application of $\mathcal{T}$. Crucially, we argued that the interaction between these operators and negation is governed by the ontological distinction between states and events following Kusumoto 2011. This necessitates a scopal distinction: negation applies low for stative predicates, preserving the stative structure required for inchoative coercion, but high for eventive predicates. This architecture successfully derives the semantic vacuity of negation in achievements, accomplishments, and activities embedded under *yiqian*, where the $\mathcal{T}$ operator targets the identical boundary point in both the positive and negative variants of an embedded clause. Further, our proposed formal mechanisms correctly predicted that the interpretation of negation in embedded statives under *yiqian* alters the truth conditions, and embedded statives under *yiqian* are forced to be resultant states as a direct constraint posed by inchoative coercion. Finally, we successfully derived the infelicity of embedded imperfectives under *yiqian* via the failure of the uniqueness presupposition involved in the $\mathcal{T}$ operator.

Furthermore, our analysis sheds light on the asymmetry between *yiqian* ‘before’ and *yihou* ‘after’. We showed that while *yiqian* selects our proposed $\mathcal{T}$ operator, *yihou* behaves differently, selecting the earliest operator. This simple selectional difference explains why *yihou* does not license semantically vacuous negation like *yiqian* does and introduces future research questions, for example, why *yiqian* and *yihou* seem to be sensitive to different types of temporal junctures.

Finally, this study has broader methodological implications for the cross-linguistic study of ‘expletive negation.’ We claim that ‘expletive negation’ is merely a descriptive label for the surface phenomenon where negation appears to be semantically vacuous in specific contexts, but it does not define a natural class of puzzles that call for a unified analysis from a theoretical perspective. The claim is based on two observations. On the one hand, the same ‘expletive negation’ trigger has vastly different behaviours in different languages. On the other hand, different ‘expletive negation’ triggers in the same language also exhibit different distributions.

As an example of the former observation, we have seen in the current study that negation appears semantically vacuous under ‘before’ in languages like Catalan, Italian, German, and Mandarin. However, in Romance languages, the phenomenon is constrained by non-veridicality (Espinal 2000, Del Prete 2008); in German, it requires matrix negation (Krifka 2012). In contrast, we have shown that Mandarin *yiqian* requires a strictly compositional, temporal-aspectual account.

To illustrate the latter observation, we compare *yiqian* with a class of other expletive negation triggers in Mandarin, including *chadian* ‘almost’ (discussed extensively in Kaufmann & Xu 2015), *haipa* ‘to fear’, *houhui* ‘to regret’, etc. One striking difference between this class of triggers and *yiqian* is that with this class of expletive negation triggers, whether the embedded negation is

expletive depends on the speaker's (or, with the attitude verb triggers, the attitude holder's) bias toward the embedded proposition. An expletive negation can be optionally inserted when the embedded proposition is undesirable from the perspective of the speaker. However, when the embedded proposition is desirable, negating the embedded proposition reverses its polarity. The following examples illustrate the data pattern with negation embedded under *chadian* 'almost':

- (50) a. wo chadian shuaidao.
1.SG almost fall.to.the.ground
'I almost fell to the ground.'
- b. wo chadian mei shuaidao.
1.SG almost NEG fall.to.the.ground
'I almost fell to the ground.'
- (51) a. wo chadian tongguo kaoshi.
1.SG almost pass exam
'I almost passed the exam.'
- b. wo chadian mei tongguo kaoshi.
1.SG almost NEG pass exam
'I passed the exam (with a small margin).'

Suppose that 'almost' scopes over the whole clause. In (50), the embedded proposition 'I fell to the ground' is clearly undesirable for the speaker, and inserting a negation does not change the meaning of the sentence (i.e., negation appears to be expletive). In contrast, in (51), the embedded proposition 'I passed the exam' is typically a desirable result for the speaker, and inserting a negation reverses the polarity of the sentence (i.e., negation is clearly interpreted). Our compositional, temporal-aspectual approach for interpreting negation under *yiqian* clearly cannot be extended to the class of expletive negation triggers represented by *chadian*, as on the one hand, *chadian* is not a temporal connective like *yiqian*; on the other hand, the interpretation of negation embedded under *yiqian* is never dependent on speaker biases.

In sum, the two observations we discussed above underscore the necessity of looking beyond the descriptive label of 'expletive negation.' The current study has worked toward a more fine-grained understanding of the interplay between negation, aspect, and temporal ordering; a natural next step for future research is to investigate the interaction between negation and speaker bias, aiming to theorize the pragmatically dependent expletive negation triggers like *chadian* 'almost' in Mandarin.

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